# Aeolus Renewables

Drivetrain condition assessment using 10-minute SCADA measurements

**131,760**
sensor records

**15**
turbines

**3.00%**
recorded faults

Predictive Modeling with Machine Learning and Neural Networks | Low-Code Path



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02 Exploratory analysis and preprocessing

03 Model comparison and final selection

04 Actionable recommendations

A Appendix: data, baselines, tuning and evidence





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# Executive Summary

| VALIDATION RECALL | TEST RECALL | TEST PRECISION | TEST F1 |
|---|---|---|---|
| **92.49%** | **86.40%** | **49.48%** | **62.93%** |
| Tuned ANN selected | 1,245 / 1,441 faults | False-alert tradeoff | Balance of P and R |

The operational decision

## Use the ANN as analyst decision support, not an automatic turbine shutdown trigger.

1,271 false-positive records require remote triage; 196 fault records were missed.



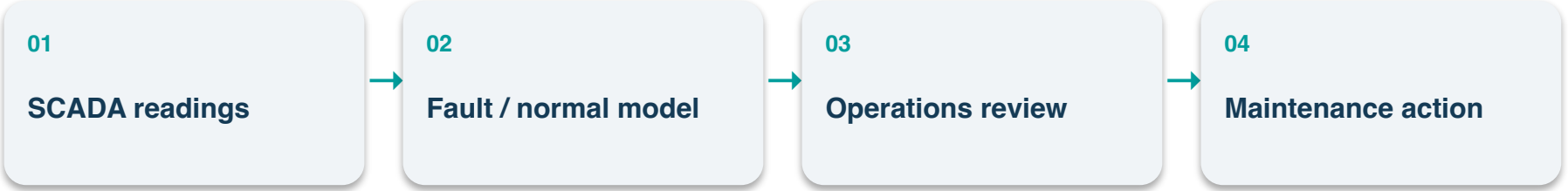
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# Business Problem Overview and Solution Approach

## WHY IT MATTERS

**Missed drivetrain faults can lead to extended downtime, emergency repairs and lost energy production.**

## CONDITION-BASED WORKFLOW



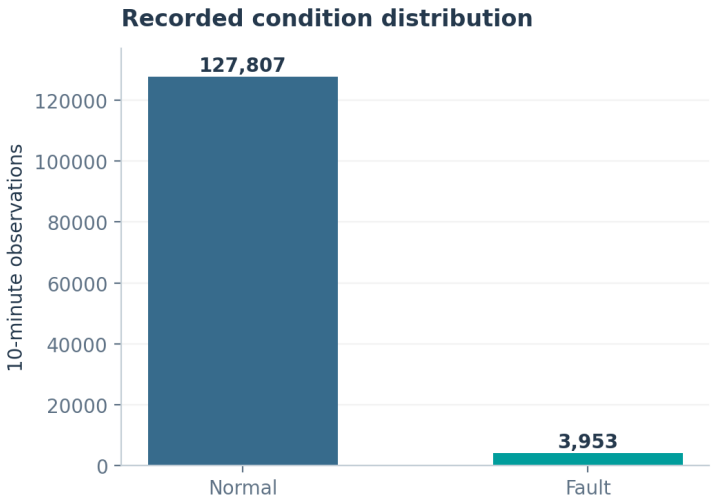

Method: timestamp-grouped chronological split, training-only preprocessing, five baseline models, two tuned candidates, validation-driven selection.

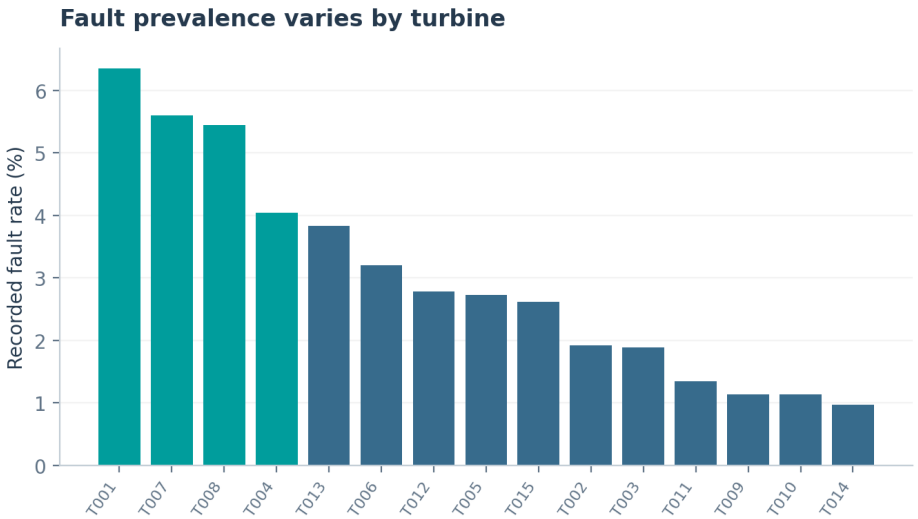


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# EDA Results — Class Balance and Asset Patterns

**131,760 readings • 15 turbines • 3.00% recorded faults**



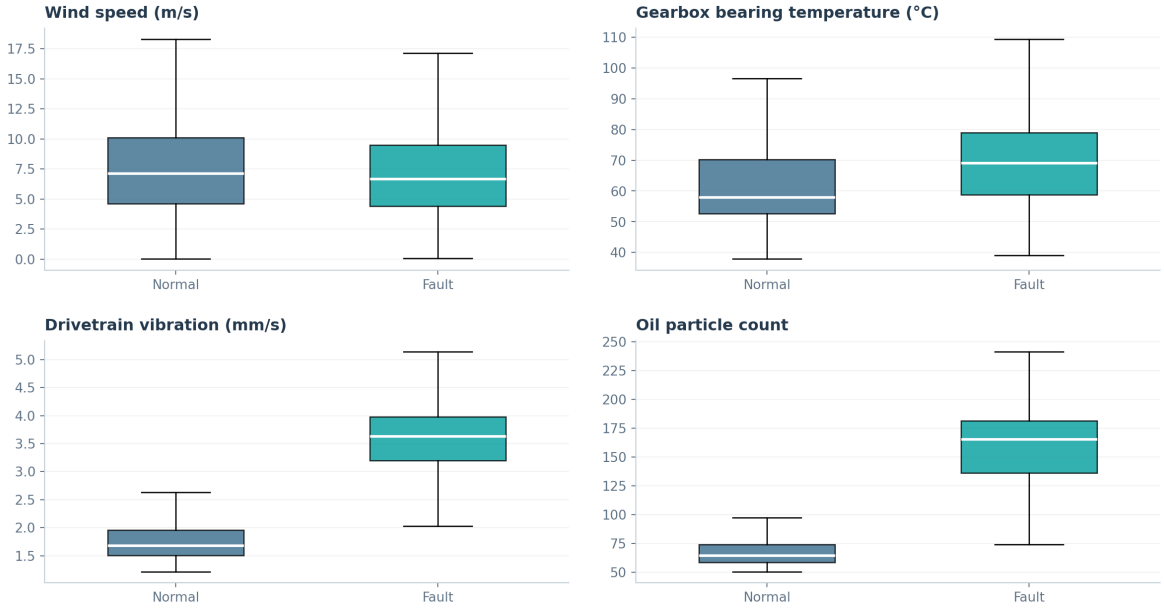




The minority fault class makes accuracy alone misleading; turbine-level rates support prioritised investigation.



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# EDA Results — Sensor Evidence




## KEY OBSERVATIONS

Vibration, gearbox temperature and oil particles are visibly elevated in fault-state records.

Wind speed alone provides weaker separation, supporting multi-signal evaluation.



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# Data Preprocessing and Leakage Prevention

### DATA QUALITY

0 duplicate rows; missing gearbox-oil (645), generator-bearing (690), oil-pressure (636).

### TREATMENT

Training-median imputation; ANN StandardScaler fit using training rows only.

### FEATURE ENGINEERING

Time cycles, power utilisation and temperature-rise variables.

### EVALUATION DESIGN

Timestamp-grouped chronological 70% / 15% / 15%; test reserved until final evaluation.



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# Model Performance Summary and Final Selection

| Tuned model | Validation recall | F1 | Precision |
|---|---|---|---|
| ANN (selected) | 92.49% | 53.54% | 37.67% |
| XGBoost | 69.10% | 69.34% | 69.57% |

**Selection rule**

**Maximise validation recall; F1 and precision are secondary tie-breakers.**

ANN catches more recorded fault states. XGBoost offers better precision and F1, so the false-alert tradeoff must be managed.



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# Actionable Business Recommendations

### 01 Triage

Review persistent model alerts and corroborating sensor signals remotely.

### 02 Dispatch

Prioritise technician inspections when risk persists across successive intervals.

### 03 Learn

Link alerts to work orders; track false alarms, missed faults, and operational workload.

### 04 Validate

Run prospective field validation before promising lead time or cost savings.



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## SCADA panel

8,784 timestamp groups, recorded at 10-minute intervals across 15 turbines.

**Environment**

Wind, ambient conditions

**Operation**

Power and rotational speeds

**Condition**

Bearing heat, vibration, oil particles

**Asset history**

Hours, load cycles and maintenance

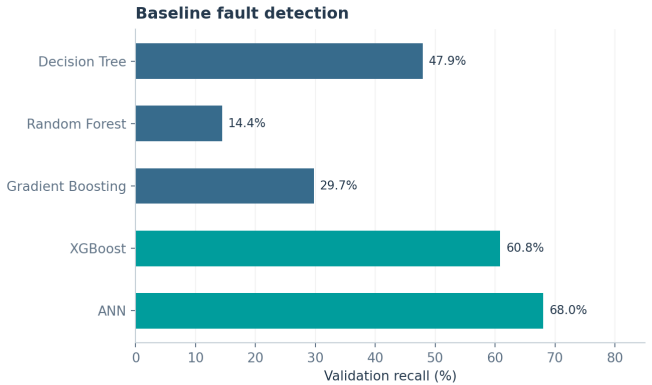
**Target: failure = 1 denotes a recorded fault condition, not a future failure horizon.**



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# Model Building — Five Baseline Classifiers

| Model | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| Decision Tree | 90.70% | 27.74% | 47.93% | 35.14% |
| Random Forest | 95.35% | 82.87% | 14.44% | 24.59% |
| Gradient Boosting | 94.54% | 46.96% | 29.74% | 36.42% |
| XGBoost | 96.64% | 71.01% | 60.83% | 65.53% |
| ANN | 94.79% | 50.28% | 68.05% | 57.83% |

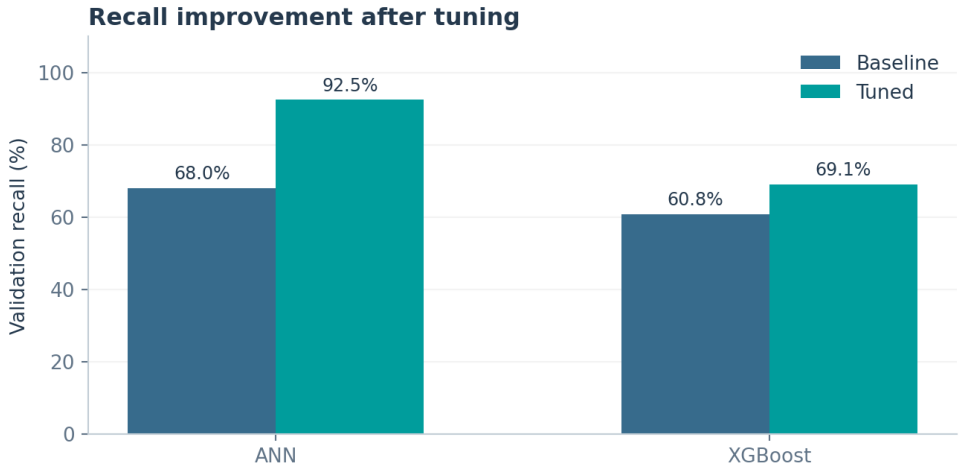



**ANN and XGBoost have the highest baseline recall and were selected for tuning.**



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# Model Improvement — Hyperparameter Tuning




## XGBOOST

6 sampled configurations; 2 timestamp-grouped expanding folds; recall objective.

Best CV recall 48.45%; holdout validation recall 69.10%.

## ANN

64/24 hidden units; BatchNorm, dropout 0.30/0.15, Adam, early stopping.



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# Final Model Performance — Untouched Test

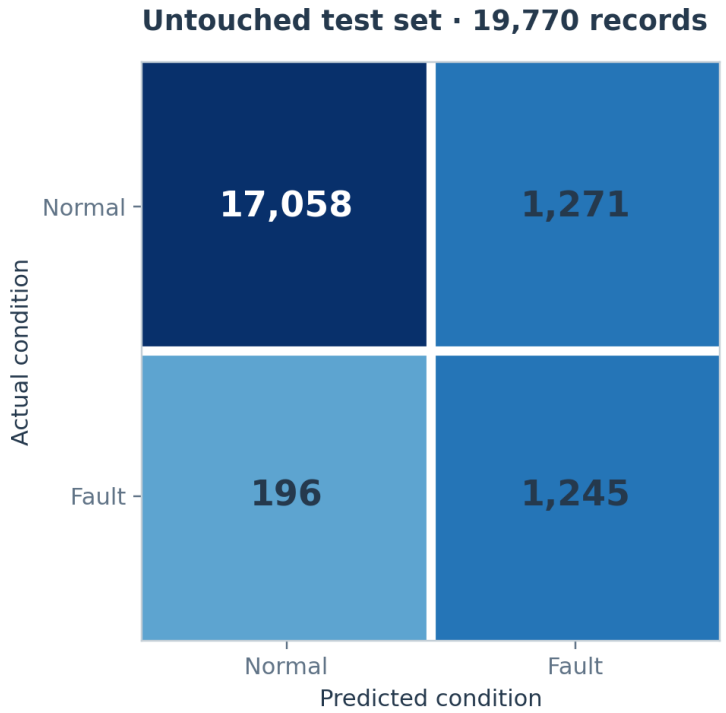
TEST RECALL
**86.40%**

TEST PRECISION
**49.48%**

TEST ACCURACY
**92.58%**

TEST F1
**62.93%**

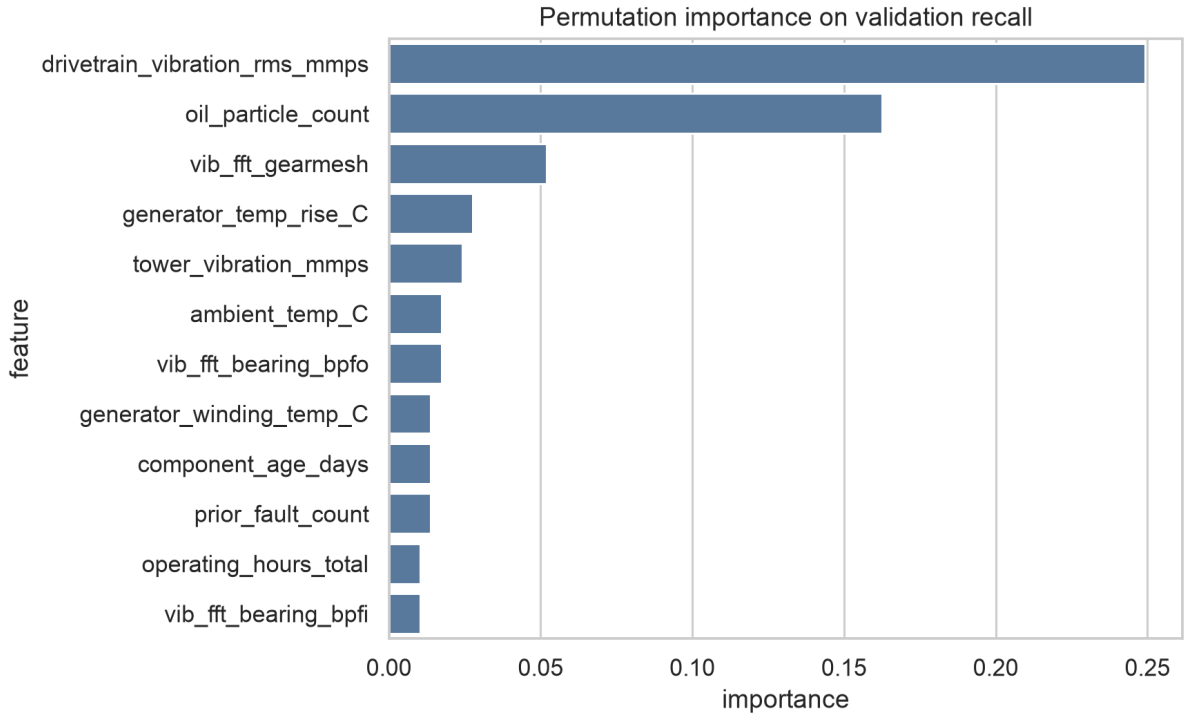
**1,245 detected • 196 missed**
**1,271 false alerts • 17,058 true normals**






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# Feature Interpretation and Limitations




**KEY DRIVERS**

Drivetrain vibration and oil particle count show the largest validation-recall impacts.

Permutation importance measures association, not causation or a future-failure lead time.



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# Timestamp-Grouped Split Verification

**TRAIN**

**92,220**

6,148 timestamps

**VALIDATION**

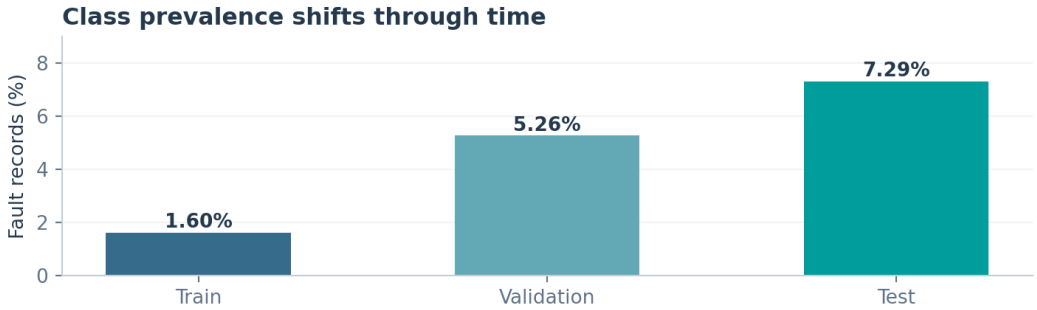
**19,770**

1,318 timestamps

**TEST**

**19,770**

1,318 timestamps




**LEAKAGE CHECK**

All 15 turbine records at a timestamp remain in the same partition. Failure prevalence changes over time.



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# Operational Use Case and Next Steps

### STAGE 1

Model flags current abnormal condition.

### STAGE 2

Analyst checks persistence and supporting sensors.

### STAGE 3

Maintenance team prioritises planned inspection.

### DEPLOYMENT GUARDRAILS

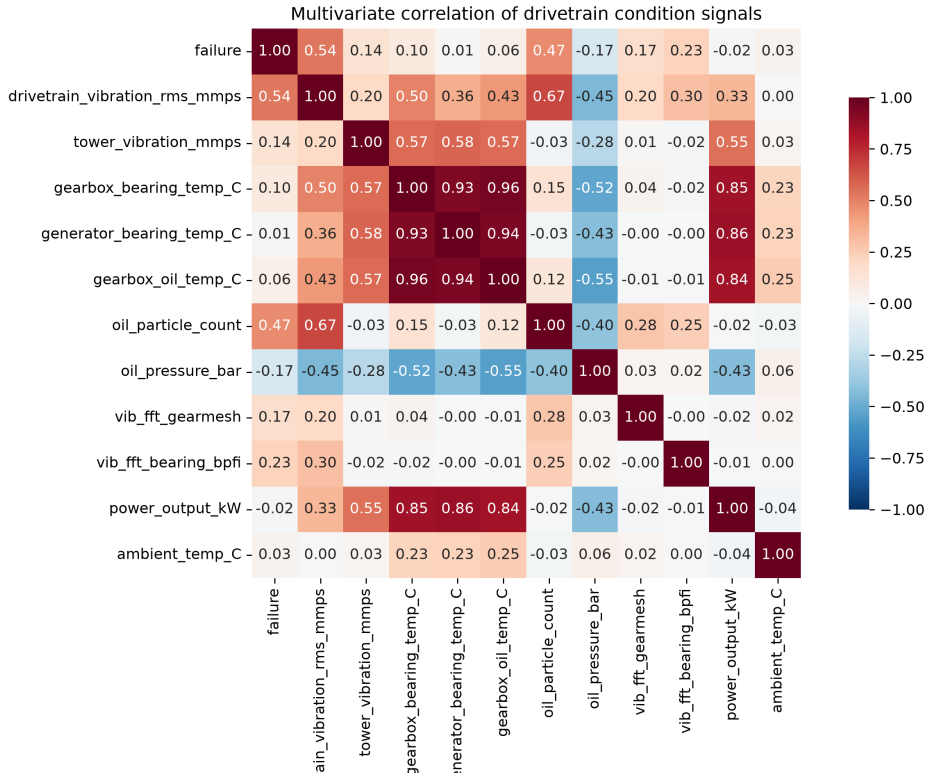
Retrospective proof of concept: prospectively validate alert value, technician workload, model drift and downtime outcomes before automated action or savings claims.



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# EDA Results — Multivariate Sensor Relationships

Correlation heatmap uses actual SCADA values across thermal, vibration, oil and operating variables.




## What the data shows

- Drivetrain vibration has the s
correlation with recorded fault
- Oil-particle count is also mat
bearing-frequency vibration is
- Oil pressure is negatively rel
bearing temperature is weake

## Business implication

No single sensor separates ev
channel model combines com
and reduces dependence on r
thresholds.

tion prohibited.

# Tuning Comparison — Recall First, Precision Trade-

Both candidates were chosen from baseline validation evidence. Selection used holdout valida
results were excluded.

| Model | Baseline P / R / F1 | Tuned P / R / F1 | |
|---|---|---|---|
| ANN | 50.28% / 68.05% / 57.83% | 37.67% / 92.49% / 53.54% | |
| XGBoost | 71.01% / 60.83% / 65.53% | 69.57% / 69.10% / 69.34% | |

e

Why ANN is selected
ANN has the highest validation recall (92.49%), so it detects more recorded fault states
business priority. XGBoost has higher precision and F1, making it a credible lower-alert-
This trade-off is documented rather than hidden.

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